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Model of superconducting alternating current bolometers

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We present a model of the response of superconducting AC bolometers to signal and noise using parameters and matrix formalism that were previously adopted in modeling DC bolometers and microcalorimeters. The model was developed to investigate novel AC bolometer designs, which have properties not described by previous models. It is applicable to kinetic inductance devices, kinetic inductance thermometers, AC transition edge sensors, and other technologies based on AC read out and frequency mixing. The AC bolometer model includes the temperature and current dependence of both resistive and inductive bolometers and microcalorimeters. The bolometers may be thermal or athermal. The model compactly describes the complex relationships between electrical and thermal signals and it simplifies the associated calculations. This model is useful for characterizing bolometers, predicting the effects of various noise sources on performance and designing new detectors. Additionally, a measurement technique is developed for characterizing these devices. © *2013 American Institute of Physics*. [http://dx.doi.org/10.1063/1.4790146]

I. INTRODUCTION

The AC bolometer model of this paper was developed primarily to investigate novel technologies such as new designs of kinetic inductance devices[1–3] (KIDs) that operate like thermal bolometers or transition edge sensors[4,5] (TESs) with GHz AC read out, but it also elucidates the physics of existing AC bolometers. Preexisting models[1,6] are insufficient for many of these applications because they do not simultaneously include many of the effects that are relevant: Bolometer impedance depends on current and temperature. Electrical and thermal signals with different frequencies are coupled to each other through varying dissipation and reactance, resulting in electrothermal feedback. Various forms of noise are included. The electrical sidebands vary independently. And, the signals are demodulated to be read out. The model distills these complex relationships into simple 3×3 matrices and provides a method for computing the behavior of the system, which has not been available prior to this work. The mathematical techniques presented here are also applicable to the development of other kinds of technologies involving the mixing of signal and noise at different frequencies, such as other types of AC bolometers, mixers, or nonlinear optics.

In Sec. I, we describe a thermal model of AC bolometers and how it may be extended to athermal devices. In Sec. II, the set of equations that describe the electrical and thermal behavior of the model are presented, neglecting the current dependence of the bolometer impedance. These equations are linearized, using a small signal approximation and represented in the matrix formalism. In Sec. III, a change of variables is used to describe demodulation. The demodulated signals are applied in Sec. IV to calculate effects of current dependent impedance. In Secs. V–IX, predictions are derived from the model, such as the signal and noise of the bolometer for various measurements. In the final section, we describe how to extend the model to include the effects of stray impedance, biasing off of resonance, and additional thermal couplings.

A. Superconducting bolometers and microcalorimeters

Bolometers[7] are designed to measure very small amounts of power. Power radiated on to a superconducting bolometer changes its electrical impedance, which is read out by amplifiers and other electronics, including components at low temperature and room temperature. The bolometer is thermally coupled to a cold bath, which keeps the bolometer near its steady state operating temperature. Microcalorimeters[8] are similar, but they are designed to measure pulses created when particles such as individual photons are absorbed. In this paper, we refer to bolometers but the results are applicable to microcalorimeters as well. These detectors are operated at low temperature and designed to have small heat capacity to minimize thermal noise.

A superconducting bolometer can be primarily resistive, such as a TES, which is operated in the phase transition between the superconducting and normal states. Or it may be primarily inductive, such as a KID or a kinetic inductance thermometer (KIT),[9] which are both operated in the superconducting state. An AC bolometer is biased with an alternating current or voltage and the thermal response of the bolometer, which is in a low frequency band, is modulated up by the AC bias to a band centered on the carrier frequency ω_0. The response of the bolometer is read out to find the power deposited in the absorber.

Arrays of AC bolometers are multiplexed[10,11] by embedding each bolometer in a resonance circuit with a unique resonant frequency ω_0 and each bolometer is biased at resonance. The frequency bands assigned to the bolometers are separated to prevent the bolometers from interacting with one another. With this set-up, the combined signals from many pixels may be read out on a single pair of wires. The signals are demodulated to determine the signal from each of the bolometers.

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B. Thermal bolometer model

The bolometer model is described in terms of thermal bolometers. Adaptions of the model to athermal bolometers will be discussed in Sec. I C. Fig. 1 shows a schematic representation of the model. The absorber, which receives the radiation to be measured, is assumed to be thermally well coupled to the bolometer. (In Sec. XI, we briefly describe how to treat more complicated bolometers.) The internal energy of the bolometer is represented by an effective temperature T and thermal quantities are used to describe the state of the bolometer. The bolometer has a heat capacity C. Heat in the bolometer is generated by absorption of radiation, represented by power P_A and by Joule heating from the current I through the bolometer resistance R. Heat flows out from the bolometer to the bath at temperature T_b as described[5] by the cooling power $P = K(T^N - T_b{}^N)$. The steady state temperature of the bolometer is T_0, and the thermal coupling is $G = \partial P/\partial T|_{T_0}$. The bolometer temperature varies in response to signal and noise. Additionally, thermal fluctuation noise (TFN) in the internal energy of the bolometer, often referred to as phonon noise, is represented by power P_{TFN}. Johnson noise and other electronic noise, which cause a fluctuation in the voltage across the thermometer, are represented by voltage V_e in series with the thermometer. Electronic noise within the thermometer causes the dissipation in the thermometer to fluctuate by $-IV_e$.

When energy is absorbed by a bolometer, the temperature is increased, which in turn affects its impedance. The voltage or current in the superconducting bolometer is measured to obtain the signal. Additionally, the impedance of a superconducting device is affected by electric current through it due to the magnetic field that the current generates. The AC bolometer model accounts for the first order effects of AC modulation, electrical components, thermal couplings, and electrothermal feedback.

C. Athermal bolometers

In the previous paragraphs, the bolometer was described in terms of thermal quantities illustrated in Fig. 1, such as heat capacity and thermal couplings. With some modifications, the model can also be applied to devices with athermal components. The athermal physics requires a microscopic model of the excitations. Given a microscopic model, the equivalent thermal couplings and noise are defined to represent the system so that the AC bolometer model describes the linear response to signals and fluctuations in an athermal bolometer. As in the thermal case, the ensemble average is used to define an effective temperature and heat capacity of the bolometer in the neighborhood of the operating point. The relaxation time of the excitations is represented by the ratio C/G and statistical variations in the excitations are represented as noise. The variations may contribute to the internal energy fluctuations P_{TFN} or the electronic noise fluctuations V_e.

II. SMALL SIGNAL ANALYSIS

In small signal analysis, sources of signal and noise are assumed to be small. Though current swings associated with the AC carrier are large, the displacements in the carrier′s amplitude and phase away from the steady state response are small because the displacements result from small sources. Similarly, displacements in the temperature are also small. In this analysis, only terms of zero and first order are significant. Fig. 2 has an illustration of the thermal and electrical signals. To first order, the modulation couples thermal signals in the thermal band (TB) with frequency ω to electrical signals with frequency $\omega_0 + \omega$ in the upper side band (USB) and frequency $\omega_0 - \omega$ in the lower side band (LSB). It is noted that electrical signals at frequency $\omega_0 + \omega$ and frequency $\omega_0 - \omega$ also couple to frequencies ω and $2\omega_0 \pm \omega$ in the thermal circuit. However, the response at frequencies $2\omega_0 \pm \omega$ is far outside the thermal bandwidth of the bolometer, so they are negligible.

We first consider a bolometer in which the impedance depends on the effective temperature, but not on the current. We will include the current dependence in Sec. IV. The first steps will be expressed in cosine and sine functions in order to multiply components of different frequencies. When the multiplication is completed, phasor notation will be used to simplify linear computations.

In the small signal approximation, the bolometer response is of the form

$$\begin{aligned} T(t) &= T_0 + T_{TB}\cos(\omega t + \theta) \quad \text{and} \\ I(t) &= I_0\cos(\omega_0 t) + I_{USB}\cos((\omega_0 + \omega)t + \phi_{USB}) \\ &\quad + I_{LSB}\cos((\omega_0 - \omega)t + \phi_{LSB}). \end{aligned} \tag{1}$$

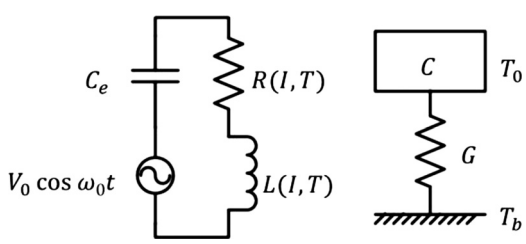

FIG. 1. *The electrical (left) and thermal (right) circuits of the AC bolometer model*. The bolometer has an impedance such that the resistance R and inductance L may vary, depending on both temperature T and current I. The bolometer impedance together with the capacitance C_e in the bias circuit determine the resonant frequency ω_0 of the resonator. The bolometer is biased by an AC voltage at resonance. The bolometer has heat capacity C at temperature T_0 and it is coupled to the bath temperature T_b through thermal coupling G.

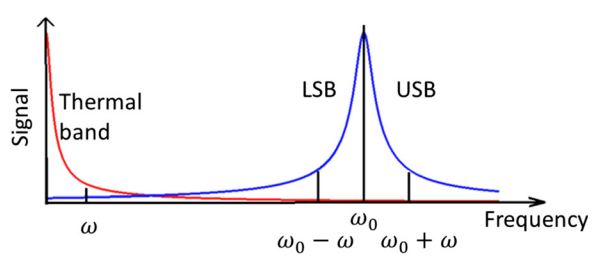

FIG. 2. The electrical and thermal bands of the AC bolometer model. The TB lies near DC. The AC bias couples thermal signals at frequency ω to electrical signals at frequency $\omega_0 + \omega$ in the USB and frequency $\omega_0 - \omega$ in the LSB. The USB and LSB comprise the electrical band, which is centered at frequency ω_0. Typically the electrical bandwidth is larger than the thermal bandwidth so that thermal signals are not strongly attenuated by the resonant circuit.

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The coefficients T_0 and I_0 are constants representing the steady state solution. Coefficients $T_{\rm TB}$, $I_{\rm USB}$, and $I_{\rm LSB}$ give the magnitudes of small displacements from the steady state due to noise and small signals. The phases of the displacements are θ, $\phi_{\rm USB}$, and $\phi_{\rm LSB}$.

A. Thermal equation

We neglect the current dependent $\partial R/\partial I$ term for now. The resistance is approximated by $R_0 + (\partial R/\partial T)T$. In this case, the differential equation, corresponding to the thermal circuit in Fig. 1, is

$$C(T)\dot{T} + GT - I^2 R_0\left(1 + \frac{1}{R_0}\frac{\partial R}{\partial T}T\right) = -IV_{\rm e} + P_{\rm TFN} + P_{\rm A}. \quad (2)$$

We substitute for I and T using Eq. (1) and retain only terms at frequency ω. The left hand side of Eq. (2) evaluates to

$$\begin{aligned}&-I_0 I_{\rm USB} R_0 \cos(\omega t + \phi_{USB}) - I_0 I_{\rm LSB} R_0 \cos(\omega t - \phi_{LSB})\\ &\quad + \omega C T_{\rm TB}\sin(\omega t + \theta) + G T_{\rm TB}\cos(\omega t + \theta)\\ &\quad - \frac{1}{2}{I_0}^2\frac{dR}{dT}T_{\rm TB}\cos(\omega t + \theta). \end{aligned} \quad (3)$$

Near equilibrium, the first order terms in $C(T)$ cancel out in Eq. (3), so the heat capacity is represented by a constant C. With the multiplication completed, phasor notation will now be adopted, which simplifies the following linear computations. In phasor notation, $T_{\rm TB}$ becomes a complex number $\boldsymbol{T}_{\rm TB} = T_{\rm TB}\exp(i\theta)$ and the actual temperature is $T = T_0 + {\rm Re}[\boldsymbol{T}_{\rm TB}\exp(i\omega t)]$. Similarly, $\boldsymbol{I}_{\rm USB} = I_{\rm USB}\exp(i\phi_{\rm USB})$ and $\boldsymbol{I}_{\rm LSB} = I_{LSB}\exp(i\phi_{LSB})$. (The phasors are represented using boldface symbols.)

Equations (2) and (3) are written in phasor notation as

$$\begin{aligned}&-I_0\boldsymbol{I}_{\rm USB}R_0 - I_0{\boldsymbol{I}_{\rm LSD}}^* R_0 + \left(i\omega C + G - \frac{1}{2}{I_0}^2\frac{dR}{dT}\right)\boldsymbol{T}_{\rm TB}\\ &\quad = \frac{-I_0(\boldsymbol{V}_{\rm e}(\omega_0+\omega) + \boldsymbol{V}_{\rm e}(\omega_0-\omega)^*)}{2} + \boldsymbol{P}_{\rm TFN}(\omega)\\ &\quad + \boldsymbol{P}_{\rm A}(\omega). \end{aligned} \quad (4)$$

B. Electrical equation

Although the resonator may be embedded in a complex circuit, it can be represented by an equivalent LCR resonator, as depicted in Fig. 1. The total voltage around the loop is given by the sum of the voltages of the inductor, resistor, capacitor, and by the bias $V\cos(\omega_0 t)$ and electronic noise V_e

$$V_{\rm L} + V_{\rm R} + V_C = V\cos(\omega_0 t) + V_{\rm e}. \quad (5)$$

In superconducting bolometers, the inductance can vary with temperature T and current I in the device. The voltage of a nonlinear inductor L is $d(LI)/dt$. By differentiating with respect to time, the voltage drop across the inductance becomes

$$V_{\rm L} = \frac{d\Phi}{dt} = \frac{d(LI)}{dt} = L\dot{I} + \frac{\partial L}{\partial I}\dot{I}I + \frac{\partial L}{\partial T}\dot{T}I. \quad (6)$$

Neglecting the current dependence for now, the voltage across the inductance is

$$V_{\rm L} = \dot{T}I\frac{\partial L}{\partial T} + L\dot{I}, \quad (7)$$

where the inductance is approximated by

$$L = L_0 + \frac{\partial L}{\partial T}(T - T_0). \quad (8)$$

In the following, the focus is put on the USB terms only. The terms for the LSB are obtained in a similar manner. Again, the equations are expressed using cosine and sine functions to accommodate terms that mix frequencies. Equation (1) and (8) are substituted into Eq. (7) and only the terms in the USB are kept to find that the voltage across the inductance at frequency $\omega + \omega_0$ is

$$\begin{aligned}V_{\rm L}(\omega_0+\omega) = &-\frac{\omega_0+\omega}{2}\frac{\partial L}{\partial T}T_{\rm TB}I_0\sin((\omega_0+\omega)t+\theta)\\ &-(\omega_0+\omega)L_0 I_{\rm USB}\sin((\omega_0+\omega)t+\phi_{\rm USB}). \end{aligned} \quad (9)$$

In phasor notation, this voltage across the inductance is represented as

$$\begin{aligned}&V_{\rm L}(\omega_0+\omega)\\ &\quad = {\rm Re}\left[i\left(\frac{\omega_0+\omega}{2}\frac{\partial L}{\partial T}\boldsymbol{T}_{\rm TB}I_0 + (\omega_0+\omega)L_0\boldsymbol{I}_{\rm USB}\right)e^{i(\omega_0+\omega)t}\right]. \end{aligned} \quad (10)$$

And in a similar manner, the voltage across the resistance is represented as

$$V_{\rm R}(\omega_0+\omega) = {\rm Re}\left[\left(\frac{I_0}{2}\frac{\partial R}{\partial T}\boldsymbol{T}_{\rm TB} + \boldsymbol{I}_{\rm USB}R_0\right)e^{i(\omega_0+\omega)t}\right]. \quad (11)$$

The partial derivatives in Eqs. (10) and (11) are constants that describe the sensitivity of the bolometer at its operating point. At resonance, the voltages of the inductor and capacitor cancel. Near resonance, the voltage across the capacitor is approximated by

$$V_{\rm C}(\omega_0+\omega) \cong {\rm Re}[i(\omega-\omega_0)L_0\boldsymbol{I}_{\rm USB}e^{i(\omega_0+\omega)t}]. \quad (12)$$

Therefore, at USB frequency $\omega_0 + \omega$, the electrical circuit equation (5) becomes

$$\begin{aligned}&i\left(\frac{\omega_0+\omega}{2}\frac{\partial L}{\partial T}\boldsymbol{T}_{\rm TB}I_0 + 2\omega L_0\boldsymbol{I}_{\rm USB}\right) + \left(\frac{I_0}{2}\frac{\partial R}{\partial T}\boldsymbol{T}_{\rm TB} + \boldsymbol{I}_{\rm USB}R_0\right)\\ &\quad = \boldsymbol{V}_{\rm e}(\omega_0+\omega). \end{aligned} \quad (13)$$

Following the same steps as shown above, but for the LSB frequency $\omega_0 - \omega$, Eq. (5) becomes

$$\begin{aligned}&i\left(\frac{\omega_0-\omega}{2}\frac{\partial L}{\partial T}\boldsymbol{T}^*_{\rm TB}I_0 - 2\omega L_0\boldsymbol{I}_{\rm LSB}\right) + \left(\frac{I_0}{2}\frac{\partial R}{\partial T}\boldsymbol{T}^*_{\rm TB} + \boldsymbol{I}_{\rm LSB}R_0\right)\\ &\quad = \boldsymbol{V}_{\rm e}(\omega_0-\omega). \end{aligned} \quad (14)$$

In comparing Eqs. (14) to (13), we note that the $2i\omega L_0$ term has changed sign for the LSB frequency. Therefore, the

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steady state impedance of the bolometer circuit in the LSB is the complex conjugate of the steady state impedance of the circuit in the USB. Similarly, the multiplication of the carrier with the temperature signal causes the $\boldsymbol{T}_{\mathrm{TB}}^{*}$ term to be conjugated in the LSB equation (14). Equation (14) is complex conjugated to obtain

$$i\left(-\frac{\omega_0-\omega}{2}\frac{\partial L}{\partial T}\boldsymbol{T}_{\mathrm{TB}}I_0+2\omega_0 L_0\boldsymbol{I}_{\mathrm{LSB}}^{*}\right)+\left(\frac{I_0}{2}\frac{\partial R}{\partial T}\boldsymbol{T}_{\mathrm{TB}}+\boldsymbol{I}_{\mathrm{LSB}}^{*}R_0\right)=\boldsymbol{V}_{\mathrm{e}}(\omega_0-\omega)^{*}. \tag{15}$$

Equation (15) is more useful than Eq. (14) because it depends on $\boldsymbol{T}_{\mathrm{TB}}$, not its complex conjugate. Using Eq. (15), we will be able to apply linear algebra to find a solution.

C. Matrix representation

Equations (4), (13), and (15) are a set of coupled linear algebraic equations. These are rewritten using the matrix formulation as has been done with DC bolometers.[5,12,13] In this method, the coefficients are contained in a matrix $\mathbf{m}$, and the signal $\mathbf{S}$ and noise terms $\mathbf{N_j}$ are expressed as vectors

$$\mathbf{m}\cdot\mathbf{x}=\mathbf{S}+\sum_j \mathbf{N}_j. \tag{16}$$

Vector $\mathbf{x}$ contains phasors that represent the dynamic variables

$$\mathbf{x}=\begin{pmatrix}\boldsymbol{I}_{\mathrm{USB}}\\ \boldsymbol{I}_{\mathrm{LSB}}^{*}\\ \boldsymbol{T}_{\mathrm{TB}}\end{pmatrix}. \tag{17}$$

The coefficients of the dynamic variables appear in matrix $\mathbf{m}$

$$\mathbf{m}=\begin{pmatrix}Z_{\mathrm{eff}} & 0 & a(\omega_0+\omega)\\ 0 & Z_{\mathrm{eff}} & a(\omega_0-\omega)^{*}\\ v_0 & v_0 & G_{\mathrm{eff}}\end{pmatrix}. \tag{18}$$

The effective impedance coefficient in the matrix is

$$Z_{\mathrm{eff}}=i2\omega_0 L_0+R_0. \tag{19}$$

The effective thermal coupling coefficient is

$$G_{\mathrm{eff}}=i\omega C+G-\frac{1}{2}\frac{{I_0}^2R_0}{T_0}\mathrm{Re}[\alpha]. \tag{20}$$

The v_0 coefficient couples current to temperature through Joule heating in the bolometer

$$v_0=-I_0R_0. \tag{21}$$

The $a(\omega')$ coefficient couples temperature to electric current through changes in bolometer impedance. The imaginary part of $a(\omega')$ slightly varies across the resonance as a function of frequency $\omega'=\omega_0\pm\omega$

$$a(\omega')=\frac{I_0R_0\alpha(\omega')}{2T_0}. \tag{22}$$

In Eq. (22), the sensitivity of the superconducting bolometer to temperature is parameterized by

$$\alpha(\omega')=\frac{T_0}{R_0}\frac{\partial Z(\omega')}{\partial T}, \tag{23}$$

where the partial derivative is evaluated with the amplitude of the AC current is held constant. This definition resembles the definition of α for DC bolometers[5,7,12,13] except in this theory α is complex valued. It is defined with the resistance R_0 in the denominator, not the bolometer impedance, to avoid mixing real and imaginary parts. The imaginary parts of the a coefficient and α depend on frequency because the reactance included in Z increases with frequency. Near resonance, ω is much less than ω_0, and the reactance approximately equals the reactance at frequency ω_0. In this approximation, the matrix reduces to

$$\mathbf{m}\cong\begin{pmatrix}Z_{\mathrm{eff}} & 0 & a_0\\ 0 & Z_{\mathrm{eff}} & a_0^{*}\\ v_0 & v_0 & G_{\mathrm{eff}}\end{pmatrix}, \tag{24}$$

where $a_0=a(\omega_0)$.

The response of the bolometer to signal and noise is computed by solving Eq. (16). The solution is given by the matrix inverse $\mathbf{m}^{-1}$

$$\mathbf{x}=\mathbf{m}^{-1}\cdot\mathbf{S}+\mathbf{m}^{-1}\cdot\sum_j\mathbf{N}_j. \tag{25}$$

The signal vector $\mathbf{S}$ and noise vectors $\mathbf{N}_j$ are described in detail in Sec. VI.

III. DEMODULATION

Signals from AC bolometers are demodulated to obtain the cosine quadrature $I_{\cos}$ and the sine quadrature $Q_{\sin}$, which are defined by

$$I=I_{\cos}(t)\cos(\omega_0 t)+Q_{\sin}(t)\sin(\omega_0 t). \tag{26}$$

When using small signal analysis, only small displacements from the steady state solution $I_0\cos(\omega_0 t)$ are considered. Equation (26) becomes

$$I=(I_0+dI)\cos(\omega_0 t)+dQ\sin(\omega_0 t). \tag{27}$$

The small displacements $\boldsymbol{dI}$ and $\boldsymbol{dQ}$ are found from the USB and LSB signals and expressed as phasors in a new vector $\mathbf{y}$

$$\mathbf{y}=\begin{pmatrix}\boldsymbol{dI}\\ \boldsymbol{dQ}\\ \boldsymbol{T}_{\mathrm{TB}}\end{pmatrix}=\mathbf{d}\cdot\mathbf{x}, \tag{28}$$

where the demodulation matrix $\mathbf{d}$ converts from the sideband representation to the dI-dQ representation

$$\mathbf{d}=\begin{pmatrix}1 & 1 & 0\\ -i & i & 0\\ 0 & 0 & 1\end{pmatrix}. \tag{29}$$

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Equation (28) relates the sidebands at frequencies $\omega_0 \pm \omega$ to the demodulated signals at frequency ω. Note that for small signals, $\boldsymbol{dI}$ corresponds to displacements in amplitude and $\boldsymbol{dQ}/I_0$ corresponds to displacements in phase of the modulated signal.

Using Eq. (28), the coupled equations of the side band representation of Eq. (16) are expressed in the demodulated ($\boldsymbol{dI}$ and $\boldsymbol{dQ}$) representation as

$$\mathbf{M} \cdot \mathbf{y} = \mathbf{d} \cdot \mathbf{S} + \mathbf{d} \cdot \sum_j \mathbf{N}_j, \tag{30}$$

where the matrix is

$$\mathbf{M} = \mathbf{d} \cdot \mathbf{m} \cdot \mathbf{d}^{-1}. \tag{31}$$

The vectors $\mathbf{d} \cdot \mathbf{N}_j$ on the right side of Eq. (30) represent the noise expressed in terms of the variables $\boldsymbol{dI}$ and $\boldsymbol{dQ}$. In Sec. VII, $\mathbf{M}$ is shown to be similar to the matrix for DC bolometers. The response to signal and noise is

$$\mathbf{y} = \mathbf{D} \cdot \mathbf{S} + \mathbf{D} \cdot \sum_j \mathbf{N}_j, \tag{32}$$

where

$$\mathbf{D} = \mathbf{d} \cdot \mathbf{m}^{-1}. \tag{33}$$

The $\mathbf{D}$ matrix is used to compute demodulated signal and noise in $\boldsymbol{dI}$ and $\boldsymbol{dQ}$ given signal $\mathbf{S}$ and noise vectors $\mathbf{N}_j$, which were defined in terms of the sideband representation.

IV. AC CURRENT DEPENDENCE

In the above derivation, it was assumed that the impedance of the thermometer was independent of the amplitude of the AC current. However, the current significantly affects[5,6,12,13] the impedance of TESs and is expected to affect KIDs and KITs if the amplitude is large.

The variations in impedance of the inductor and resistor at frequency ω are a function of displacements in temperature $\boldsymbol{T_{TB}}$ and amplitude $\boldsymbol{dI}$ as described by

$$dZ = \left(\frac{\partial Z}{\partial T}\right)\boldsymbol{T_{TB}} + \left(\frac{\partial Z}{\partial I_{\cos}}\right)\boldsymbol{dI}. \tag{34}$$

We substitute in for $\boldsymbol{dI}$ using Eqs. (28) and (29) to find

$$dZ = \left(\frac{\partial Z}{\partial T}\right)\boldsymbol{T}_{\mathrm{TB}} + \left(\frac{\partial Z}{\partial I_{\cos}}\right)(\boldsymbol{I}_{\mathrm{USB}} + \boldsymbol{I}_{\mathrm{LSB}}*). \tag{35}$$

The effects of the temperature dependent $\partial Z/\partial T$ term in Eq. (35) were included in Sec. II, contributing to the coefficients a, a^*, and G_{eff} in the matrix of Eq. (18). Each of the amplitude dependent $\partial Z/\partial I_{\cos}$ terms has a similar form to the $\partial Z/\partial T$ term: They contribute additional terms (containing I_{USB} and I_{LSB}) to the thermal equation and the sideband equations. Applying the same steps done in Sec. II to the $\partial Z/\partial I_{\cos}$ terms, the resulting matrix is found to be

$$\mathbf{m} = \begin{pmatrix} Z_{\mathrm{eff}} + b(\omega_0+\omega) & b(\omega_0+\omega) & a(\omega_0+\omega) \\ b(\omega_0-\omega)^* & Z_{\mathrm{eff}} + b(\omega_0-\omega)^* & a(\omega_0-\omega)^* \\ v & v & G_{\mathrm{eff}} \end{pmatrix}. \tag{36}$$

The v coefficient couples current to temperature through Joule heating, replacing the v_0 coefficient in Eq. (18)

$$v = -IR_0(2 + \mathrm{Re}[\beta])/2. \tag{37}$$

The coefficient that couples changes in current through changes in impedance to voltage is

$$b(\omega') = \beta(\omega')R_0/2, \tag{38}$$

where the sensitivity to changes in bias amplitude is parameterized by parameter β

$$\beta(\omega') = \frac{I_0}{R_0}\frac{\partial Z(\omega')}{\partial I_{\cos}}\Big|_{T_0}. \tag{39}$$

The partial derivative represents changes of bolometer impedance with the temperature held constant. The definition of β is similar to the definition of β for DC bolometers in Refs. 5, 12, and 13, but for AC bolometers, it is complex valued. The parameter β varies with frequency due to the reactance of Z.

Near resonance,

$$\mathbf{m} \cong \begin{pmatrix} Z_{\mathrm{eff}} + b_0 & b_0 & a_0 \\ {b_0}^* & Z_{\mathrm{eff}} + {b_0}^* & {a_0}^* \\ v & v & G_{\mathrm{eff}} \end{pmatrix}. \tag{40}$$

Where $b_0 = b(\omega_0)$.

The effect of the AC current on bolometer impedance is assumed be approximately independent of frequency in the narrow range of frequencies near resonance. In this case, the direct effect of the USB and LSB current on the impedance is equal. This implies that $\boldsymbol{dQ}$ does not directly affect the impedance because $\boldsymbol{dQ}$ is associated with the difference between the two sidebands. For this reason, the impedance Z is not modeled as a function of $\boldsymbol{dQ}$. If necessary, such an effect could be modeled by including $\boldsymbol{dQ}$ along with a frequency dependent $\beta(\omega')$, which differs between the sidebands.

Under AC bias, the voltage across a superconducting bolometer may vary in a nonlinear way during each cycle of the AC bias current. The AC carrier and the induced voltage are periodic. The linear component of the voltage appears at frequency ω_0. The nonlinear response appears at multiples of ω_0. These nonlinear components of the voltage induce negligible currents because they lie far outside of the resonator bandwidth.[6] Likewise, the nonlinear components do not affect the temperature because they do not get coupled into the thermal bandwidth. As a result, the response of the resonator is approximately a sine wave of frequency ω_0, with amplitude and phase that vary slowly. Small signals induce small displacements in current and temperature from the steady state solution. Therefore, the small signal approximation accurately

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models the current and temperature of the bolometer even though the voltage across the bolometer may have large nonlinear components. The parameters α, β, R_0, and $Z_{\rm eff}$ represent the linear response, with the nonlinearities filtered out by the resonator.

V. TESTING THE MODEL AND MEASURING MODEL PARAMETERS

The responsivity is the ratio of signal divided by input power at TB frequency ω. The coefficients $m^{-1}{}_{1,3}$ and $m^{-1}{}_{2,3}$ represent the responsivities in the upper and lower sideband, as in Eq. (25). From Eq. (32), the responsivity in amplitude $\boldsymbol{dI}$ is $D_{1,3}$ and the responsivity in $\boldsymbol{dQ}$ is $D_{2,3}$. These can be fit to measurements of bolometer response to constrain model parameters.

Measurements of the ratio between the voltage applied to the bolometer and the response in current provide additional ways to test the model and characterize the bolometer. A small stimulus voltage can be applied to the electric circuit at USB frequency $\omega_0 + \omega$ and the response can be measured at the same frequency. The ratio of these determines $m^{-1}{}_{1,1}$. The response to the same stimulus can be measured in the LSB frequency $\omega_0 - \omega$. The ratio of the USB stimulus voltage to the complex conjugate of LSB current is used to measure $m^{-1}{}_{2,1}$. Similarly, a stimulus can be applied to the lower side band to measure $m^{-1}{}_{2,2}$ and $m^{-1}{}_{1,2}$. In these expressions, the LSB voltage stimulus and LSB current response must be complex conjugated to obtain the correct values for the coefficients. These measurements, when made over a range of frequencies and used in conjunction with measurements of detector I-V curves, can be applied to find model parameters in a manner similar to complex impedance measurements in DC bolometers.[13] Using model fitting, the values of α, β, and C can be extracted. A ratio of these measurements, dubbed the side band ratio (SBR) is

$$SBR(\omega) = \frac{m^{-1}{}_{2,1} m^{-1}{}_{1,2}}{m^{-1}{}_{1,1} m^{-1}{}_{2,2}}, \tag{41}$$

where $\mathbf{m}$ is the matrix from Eq. (36). Importantly, the effects of stray impedances, attenuations, and phase shifts outside the resonant circuit divide out of measurements of the SBR because the stimulus and response in both side bands appear in both the numerator and denominator. The SBR should be less susceptible to experimental artifacts than complex impedance measurements or responsivity measurements.

VI. COMPUTING THE RESPONSE TO SIGNAL AND NOISE

From Eqs. (4), (13), (15), and (16), we find the vectors for the signal and thermal fluctuation noise are

$$\mathbf{S} = \begin{pmatrix} 0 \\ 0 \\ \boldsymbol{P}_{\rm A} \end{pmatrix}, \quad \mathbf{N_{TFN}} = \begin{pmatrix} 0 \\ 0 \\ \boldsymbol{P}_{\rm TFN} \end{pmatrix}. \tag{42}$$

The signal vector $\mathbf{S}$ and thermal fluctuation noise vector $\mathbf{N_{TFN}}$ each have one nonzero component corresponding to power put into the bolometer internal energy as described in the thermal equation (4). The phasor $\boldsymbol{P}_{\rm A}$ represents the power absorbed by the bolometer. The energy exchange between the bolometer and the bath is represented by the power $\boldsymbol{P}_{\rm TFN}$. The RMS fluctuation of this noise is

$$\langle {\rm P_{TFN}}^2 \rangle^{1/2} = \sqrt{\gamma 4 k_{\rm B} T_0{}^2 G}. \tag{43}$$

The γ factor describes how large TFN is compared to equilibrium TFN. The γ factor is 1 for devices at thermal equilibrium. It can be as small as 0.5 if the bath temperature is small compared to the temperature of the bolometer.[7] For athermal devices, $\langle P_{\rm TFN}{}^2 \rangle^{1/2}$ (which may be frequency dependent) is computed from the microscopic model of the athermal physics.

From Eqs. (4), (13), (15), and (16), we find the vectors for the electronic noise of the upper and lower side bands

$$\mathbf{N_{USB}} = \begin{pmatrix} \boldsymbol{V}_{\rm e} \\ 0 \\ -I_0 \boldsymbol{V}_{\rm e}/2 \end{pmatrix} \quad \text{and} \quad \mathbf{N_{LSB}} = \begin{pmatrix} 0 \\ \boldsymbol{V}_{\rm e}^* \\ -I_0 \boldsymbol{V}_{\rm e}^*/2 \end{pmatrix}. \tag{44}$$

The electronic noise $\boldsymbol{V}_{\rm e}$ is typically flat over the narrow bandwidth of the resonator. If the electronic noise is equal to equilibrium Johnson noise then the RMS noise $\langle V_{\rm e}{}^2 \rangle^{1/2}$ is $\sqrt{4 k_{\rm B} T R}$. In practice, the electronic noise is often several times[14] higher than that value and so

$$\langle V_{\rm e}{}^2 \rangle^{1/2} = M_{\rm e} \sqrt{4 k_{\rm B} T R}, \tag{45}$$

where $M_{\rm e}$ is a multiplier. For example, first order non-equilibrium theory predicts[15] that the current dependence of the resistance in DC TESs results in $M_{\rm e} = \sqrt{1 + 2\beta}$. The device may have greater noise if driven far from equilibrium by the bias.

The USB and LSB electronic noise vectors $\mathbf{N}_{\rm USB}$ and $\mathbf{N}_{\rm LSB}$ each have two nonzero components corresponding to the electronic voltage noise in Eqs. (13) and (15) and the corresponding internal energy fluctuations in Eq. (4).

The responses in the USB, LSB, $\boldsymbol{dI}$, and $\boldsymbol{dQ}$ are computed using Eqs. (25), (32), (36), and (40). The full solution may be found using a symbolic math solver or computed numerically. Alternatively, the equation may be simplified before solving as described in Sec. VII. When computing the noise, the RMS variation in the components of $\mathbf{x}$ or $\mathbf{y}$ is found by adding the noise from different sources in quadrature.

VII. SIMPLIFYING THE EQUATIONS

A. Demodulated signals and temperature

For the demodulated representation, the matrix $\mathbf{M}$ is found from Eqs. (29), (31), and (36). Assuming $a(\omega') \cong a_0$ and $b(\omega') \cong b_0$, the matrix is

$$\mathbf{M} \cong \begin{pmatrix} Z_{\rm eff} + 2\,{\rm Re}[b_0] & 0 & 2\,{\rm Re}[a_0] \\ 2\,{\rm Im}[b_0] & Z_{\rm eff} & 2\,{\rm Im}[a_0] \\ v & 0 & G_{\rm eff} \end{pmatrix}. \tag{46}$$

The matrix $\mathbf{M}$ describes the relationships between the amplitude $\boldsymbol{dI}$, phase $\boldsymbol{dQ}/I_0$, and temperature $\boldsymbol{T_{TB}}$. In Eq. (46), the

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amplitude and temperature are independent of the imaginary parts of α and β. However, the phase depends on both the real and imaginary parts because it is coupled to the other dynamic variables through coefficients that depend on $\mathrm{Im}[\alpha]$ and $\mathrm{Im}[\beta]$.

B. Amplitude and temperature

In Eq. (46), amplitude $\boldsymbol{dI}$ and temperature $\boldsymbol{T}_{\mathrm{TB}}$ are independent of $\boldsymbol{dQ}$. Therefore, the calculation of the amplitude response can be simplified to a 2×2 matrix, with 2-vectors. We drop the components corresponding to $\boldsymbol{dQ}$ to obtain

$$\mathbf{y} = \begin{pmatrix} \boldsymbol{dI} \\ \boldsymbol{T}_{\mathrm{TB}} \end{pmatrix} \quad \text{and} \quad \mathbf{M} = \begin{pmatrix} Z_{\mathrm{eff}} + 2\mathrm{Re}[b_0] & 2\mathrm{Re}[a_0] \\ v & G_{\mathrm{eff}} \end{pmatrix}. \tag{47}$$

The definitions of a, b, G_{eff}, v, and Z_{eff} are substituted in to the matrix of Eq. (47) to find

$$\mathbf{M} = \begin{pmatrix} i2\omega_0 L_0 + R_0(1+\mathrm{Re}[\beta]) & \dfrac{I_0 R_0 \mathrm{Re}[\alpha]}{T_0} \\ -I_0 R_0(2+\mathrm{Re}[\beta])/2 & i\omega C + G - \dfrac{1}{2}\dfrac{I_0{}^2 R_0}{T_0}\mathrm{Re}[\alpha] \end{pmatrix}, \tag{48}$$

where α and β are evaluated at frequency ω_0. Note that the matrix in Eq. (48) resembles the matrix for a DC bolometer,[5,12,13] except that in the AC bolometer α and β are complex and some of the terms differ by factors of two from the DC case.

VIII. SIGNAL AND NOISE FOR AMPLITUDE READ OUT

The RMS noise is found by computing the absolute value of the noise in the frequency domain (because the RMS does not depend on phase) and then computing the expectation value of that expression. It is divided by the responsivity to find the noise equivalent power (NEP).

The NEP in the amplitude signal $\boldsymbol{dI}$ due to upper side band electronic noise is

$$\mathrm{NEP}_{\mathrm{USB}} = \left|\frac{[\mathbf{D}\cdot\mathbf{N}_{\mathbf{USB}}]_1}{\mathbf{D}_{1,3}}\right| = \left|\frac{\left(\mathbf{D}_{1,1} - \mathbf{D}_{1,3}(I_0/2)\right)V_e}{\mathbf{D}_{1,3}}\right|. \tag{49}$$

The two terms in the numerator of Eq. (49) oppose each other, resulting in a reduction of Johnson noise and other electronic noise, as occurs in DC bolometers.[4,5,7,12] To simplify the calculation, we assume $a(\omega') \cong a_0$ and $b(\omega') \cong b_0$. Equations (33), (40), (49) and the definitions of the coefficients contained in $\mathbf{m}$ are used to compute the NEP of the USB electronic noise. In the amplitude read out, the LSB noise is equal in magnitude to USB noise. (Note that they are not equal for phase read out). The NEPs of the USB electronic noise, LSB electronic noise, and TFN are added together in quadrature to find the total NEP for amplitude readout. The square of the total NEP is found to be

$$\mathrm{NEP}^2 = 2\left|\frac{(G+i\omega C)T_0}{I_0 R_0 \mathrm{Re}[\alpha]}\right|^2 \langle V_{\mathrm{e}}{}^2\rangle + \langle P_{\mathrm{TFN}}{}^2\rangle. \tag{50}$$

If the bolometer has large loop gain,[5] the $2(GT_0/I_0R_0\mathrm{Re}[\alpha])^2 \langle V_{\mathrm{e}}^2\rangle$ term can be dropped from Eq. (50) because it is small compared to the thermal fluctuation noise term $\langle P_{\mathrm{TFN}}{}^2\rangle$. In that case, the NEP is approximated by

$$\mathrm{NEP}^2 = 2\left(\frac{\omega C T_0}{\mathrm{Re}[\alpha]}\right)^2 \frac{\langle V_e{}^2\rangle}{(I_0R_0)^2} + \langle P_{\mathrm{TFN}}{}^2\rangle. \tag{51}$$

Using the assumption that the bolometer is heated far above the bath temperature, the cooling power can be approximated by $P \cong KT^N$. The thermal conductance is given by $G = KNT^{N-1}$. The power dissipated in the AC bolometer is $I_0{}^2R_0/2$, which equals the cooling power. Consequently,

$$(I_0R_0)^2 = 2GT_0R_0/N. \tag{52}$$

Substituting Eqs. (43), (45), and (52) into Eq. (51), it becomes

$$\mathrm{NEP}^2 \cong 2\left(\frac{\omega C T_0}{\mathrm{Re}[\alpha]}\right)^2 \frac{4k_{\mathrm{B}}T_0R_0}{2GT_0R_0/N} M_{\mathrm{e}}^2 + 4\gamma k_{\mathrm{B}}T^2G_0. \tag{53}$$

The optimal full-width-half-max (FWHM) energy resolution of an AC calorimeter read out in the cosine quadrature is found from the usual integral[5] of the NEP in Eq. (53) over frequency

$$\Delta E_{FWHM} = 2.35\left(\int_0^\infty \frac{4df}{NEP^2}\right)^{-1/2} = 2.35\sqrt{M_{\mathrm{e}}\frac{4k_BT^2C}{|\mathrm{Re}[\alpha]|}\sqrt{\gamma N}}, \tag{54}$$

where $\omega = 2\pi f$ is used. This is the same energy resolution as is found for a DC bolometer.[4,5,7,12] This result is applicable to amplitude readout of AC TESs, KIDs, or KITs which are embedded in a bolometer that is weakly thermally coupled to the bath so that dissipation significantly elevates the bolometer temperature.

IX. SIGNAL AND NOISE FOR PHASE READ OUT

The responsivity in the sine quadrature $S_Q(\omega)$ is given by matrix element $D_{2,3}$. Unlike in the amplitude readout, the response depends on both the real and imaginary parts of α and β.

The description for bolometers in which the phase is read out resembles the case when amplitude is read out, but with an important difference. For amplitude readout, having high temperature sensitivity α requires that $(T/R)(\partial R/\partial T)$ is large. In phase readout however, the sensitivity is enhanced by the quality factor Q of the resonance. From Eq. (23), the sensitivity in an inductive bolometer is found to be

$$\mathrm{Im}[\alpha] = Q\left(\frac{T}{L}\frac{\partial L}{\partial T}\right), \tag{55}$$

where $Q = \omega_0 L/R_0$.

For KIDs and KITs, the parameters Q, $(T/L)\partial L/\partial T$, and α are calculable using Mattis-Bardeen theory.[1] Note that in KID bolometers, a significant amount of phase noise is

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observed. An additional noise vector must be included to model the phase noise. In the sideband representation, the vector is

$$\mathbf{N}_\theta = \begin{pmatrix} \boldsymbol{V}_\theta(\omega) \\ -\boldsymbol{V}_\theta(\omega) \\ 0 \end{pmatrix}, \tag{56}$$

where $\boldsymbol{V}_\theta$ corresponds to electronic noise voltages in series with the bolometer. The **D** matrix and the phase noise vector, along with the vectors for other electronic noise and thermal fluctuation noise, can be used to compute the noise in phase readout, as in Eq. (32).

X. STRAY IMPEDANCE

If the carrier is not precisely at resonance, the inductor and capacitor do not exactly cancel at frequency ω_0. This causes an asymmetric response to thermal signals and electronic stimulus in the sidebands. Additionally, parasitic resistances can occur that are not associated with the bolometer. Both of these can be included in extra parasitic impedance Z_p. Using the matrix of Eq. (36) or (40), the impedance Z_p is added to the USB coefficient $m_{1,1}$, and Z_p^* is added to the LSB coefficient $m_{2,2}$ Note that electronic noise from electronic elements that are not thermally connected to the thermometer do not directly change the effective temperature in the bolometer, so the temperature component of electronic noise vectors from those sources is zero.

XI. DISCUSSION

A number of additional results can be derived from the above model, such as modeling the response in the time domain and finding bolometer stability requirements. The model can be easily extended to describe complex bolometers, which may have weakly coupled absorbers and other heat capacities coupled to the bolometer. These complex bolometers can be modeled by adding additional equations to the model for each thermal body, along with a thermal fluctuation noise vector for each thermal coupling.[12,16] (These thermal fluctuation noise vectors have two nonzero components, one component for each of the two bodies in the bolometer that they couple between.) More complex bolometers result in larger matrices which may be analyzed using the steps presented above. Additionally, equivalent bolometer analysis (EBA),[17,18] which has been used to derive general relationships for complex DC bolometers, can be applied to AC bolometers as well.

The superconducting AC bolometer model presented here makes the standard analytical techniques applied to DC bolometers available to AC bolometers such as KIDs, KITs, and AC TESs. This allows for a common language that spans these fields of research, and it facilitates the exchange of ideas between the researchers involved. The authors hope that the model may be used to develop new technologies lying on the boundaries between these fields, and that it will illuminate the physics of detectors already under development.

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